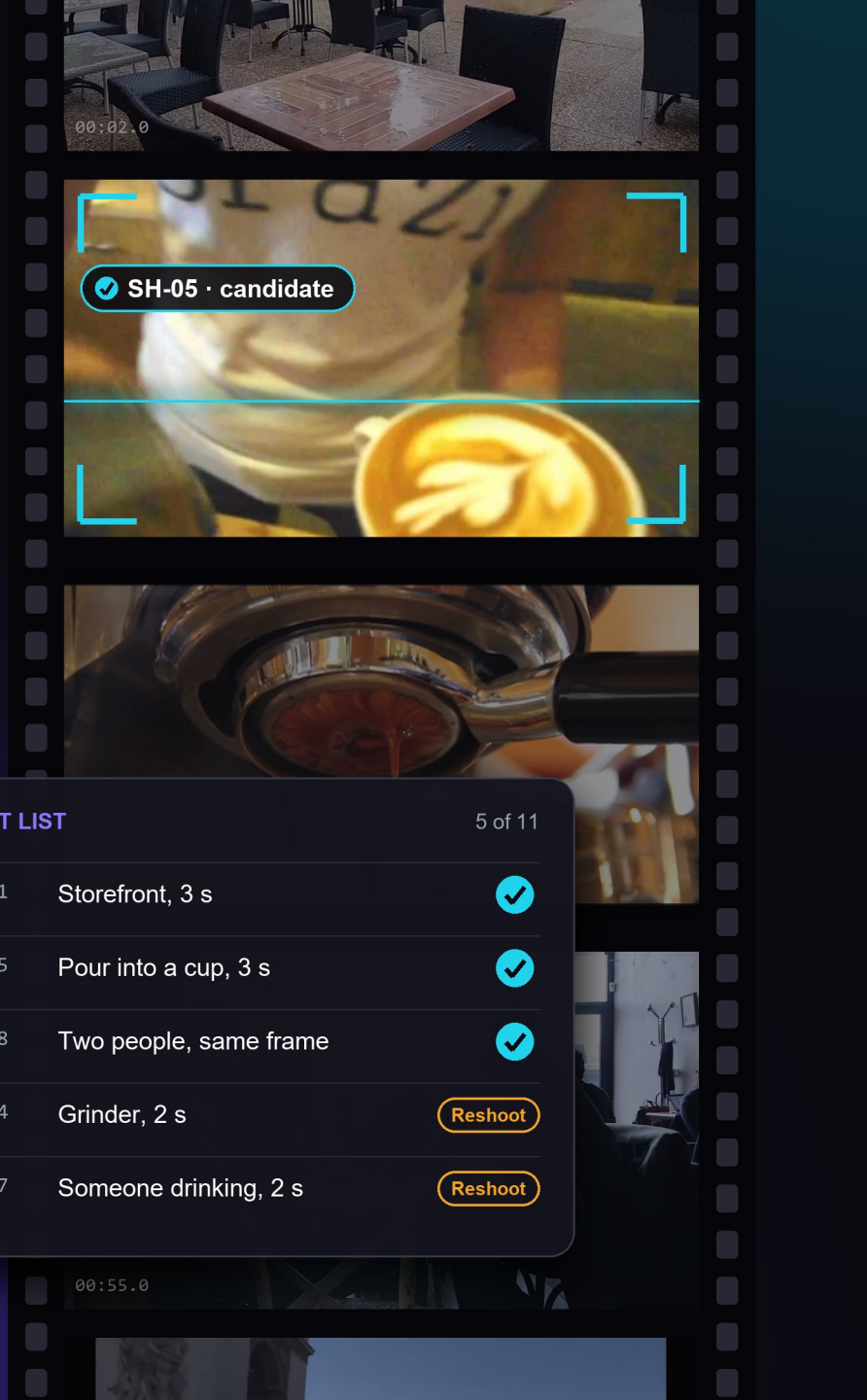
CREATOR AI · LIGHTNING TRACK · ASUS UGEN300

SceneGap Edge

Helps a crew check the shot list against the footage before leaving the location, with playable evidence and a person's decision on every shot.

- Runs on your own computer
- Evidence for every answer
- A person decides
- UGen300 port planned



Stage I proposal · rm-rf-1 · github.com/yu9bryan/scenegap-edge

PROBLEM

The missing shot is found at 23:00, after the café has closed



Illustrative scenario · two-person crew, café promo, 11 must-have shots

“We had pours. None of them was three seconds long. We found out in the edit.”

09:00	Shot list agreed with the client: storefront, interior, beans, grinder, 3 s pour, two barista takes, bag with readable brand.
17:40	Wrap. Several cards of footage; nobody has time to watch it all on site.
23:00	The editor scrubs for the pour: every take cuts away after 2 s, and nobody filmed a customer drinking.
Next week	Reshoot day: NT$12,000–28,000 per crew-day in Taiwan, US$800–1,500 per videographer-day in the US, plus café and client again.

Day rates: PRO360 quote statistics (TW), Film Lifestyle mid-level rates (US); docs/research/BUSINESS_FACTS.md.

PROBLEM

Existing tools find footage; SceneGap adds shot-list rules, evidence and reshoot tracking

TOOL	FINDS THE MOMENT IN A CLIP	SHOT-LIST RULES (LENGTH, TAKES, SAME FRAME)
Premiere Pro media intelligence	yes, In/Out range	not in official docs
DaVinci Resolve 21 IntelliSearch	visual search: whole clips	not in official docs
Final Cut Pro 12 visual search	yes	not in official docs
invideo AI script supervisor	not stated	done / pending only
SceneGap Edge	**yes, original timecode**	**checked per rule; a person decides**

> **The difference is the workflow around search: shot list → rules → evidence → a person's decision → reshoot list → re-check.**

Vendor documentation, accessed 2026-10-26 (Adobe media intelligence FAQ and others). More tools and prices: A11.

PROBLEM

What works today, and what still needs proof

Shown today (live app, real footage)

- Shot list in the crew's words, with rules: length, takes, same frame
- Candidate clips with playable evidence on the original timecode
- Rule checks: 10 of 12 controlled cases judged right (slide 12)
- A person accepts, rejects, or adds the shot to the reshoot list
- Demo: 9 Commons clips (5.9 min) analysed in 22 s on a desktop GPU; new footage re-checked file by file

Not shown yet

- Missing shots on real café video: test 4 met both targets (65% caught, 3.8% wrong reshoots), but on still photos; test 3 video had 12% wrong (slide 15)
- Time saved: no user study yet. Proxy: 317 s of candidate clips to play vs 839 s of footage (38%)
- UGen300 speed and quality: not run on the device yet
- Human-checked labels: current labels are AI-assisted

SceneGap Edge today is a review assistant that keeps evidence and a person's decision; automation grows only as it is proven.

SOLUTION

From your shot list to a decision you can play back

1 · Write the shot list

In Chinese, as the crew speaks; a person checks the English search phrase and adds rules. Test list: 11 shots, each with a minimum length; 1 same-frame, 1 two-takes, 1 manual brand check.

2 · Index footage locally

Camera cuts, one frame per second on the original timecode, CLIP vectors. Desktop: 2.3 s per minute of video (GPU), 3.1 s (CPU). UGen300 next.

3 · Check each rule

Pass, fail or can't tell per rule; failing clips stay visible. Threshold τ 0.225 set on dev; search over 10,000 segments p95 14 ms.

4 · A person decides

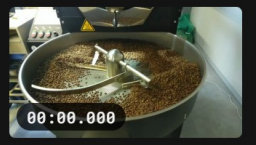
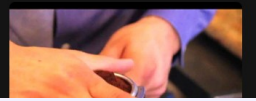
Accept, reject with a reason, or add to the reshoot list. Only the person's decision counts as done; it is bound to the shot's version.

The AI's four answers (a person's decision is stored separately)

AI ANSWER	MEANING	ON THE 26-CLIP TEST LIBRARY
Candidate	Every rule passes on at least one clip; still needs a person to accept	10 of 11 shots
Partial	Something matched but a rule failed or can't be decided (too short, one take of two, a manual check)	1 (SH-10, manual brand check)
Not found	Nothing reached the threshold in the analysed files; not proof of absence	0; 30 of 38 missing shots still got 'candidate' (slide 15)
Unverifiable	Files failed to decode or are missing, so absence cannot be claimed	0

SOLUTION

Before leaving, the crew sees which shots are confirmed

ID	鏡頭	AI 判斷	人工決定	原因	片段	
SH-01 v1	**店外街景，看得到咖啡店門面，連續 3 秒** a photo of the outside of a cafe seen from the street	AI ◎ 有候選	✓ 已接受	–	 已接受 · 00:00.000–00:10.012	看片段
SH-02 v1	**店內全景，看得到桌椅，連續 3 秒** a photo of the inside of a cafe with tables and chairs	AI ◎ 有候選	✓ 已接受	–	 已接受 · 00:55.013–00:59.017	看片段
SH-03 v1	**咖啡豆特寫，連續 2 秒** a close-up photo of roasted coffee beans	AI ◎ 有候選	○ 未確認	–	 00:00.000–00:09.000	看片段
SH-04 v1	**磨豆過程，連續 2 秒** a photo of a coffee grinder	AI ◎ 有候選	○ 未確認	–		看片段

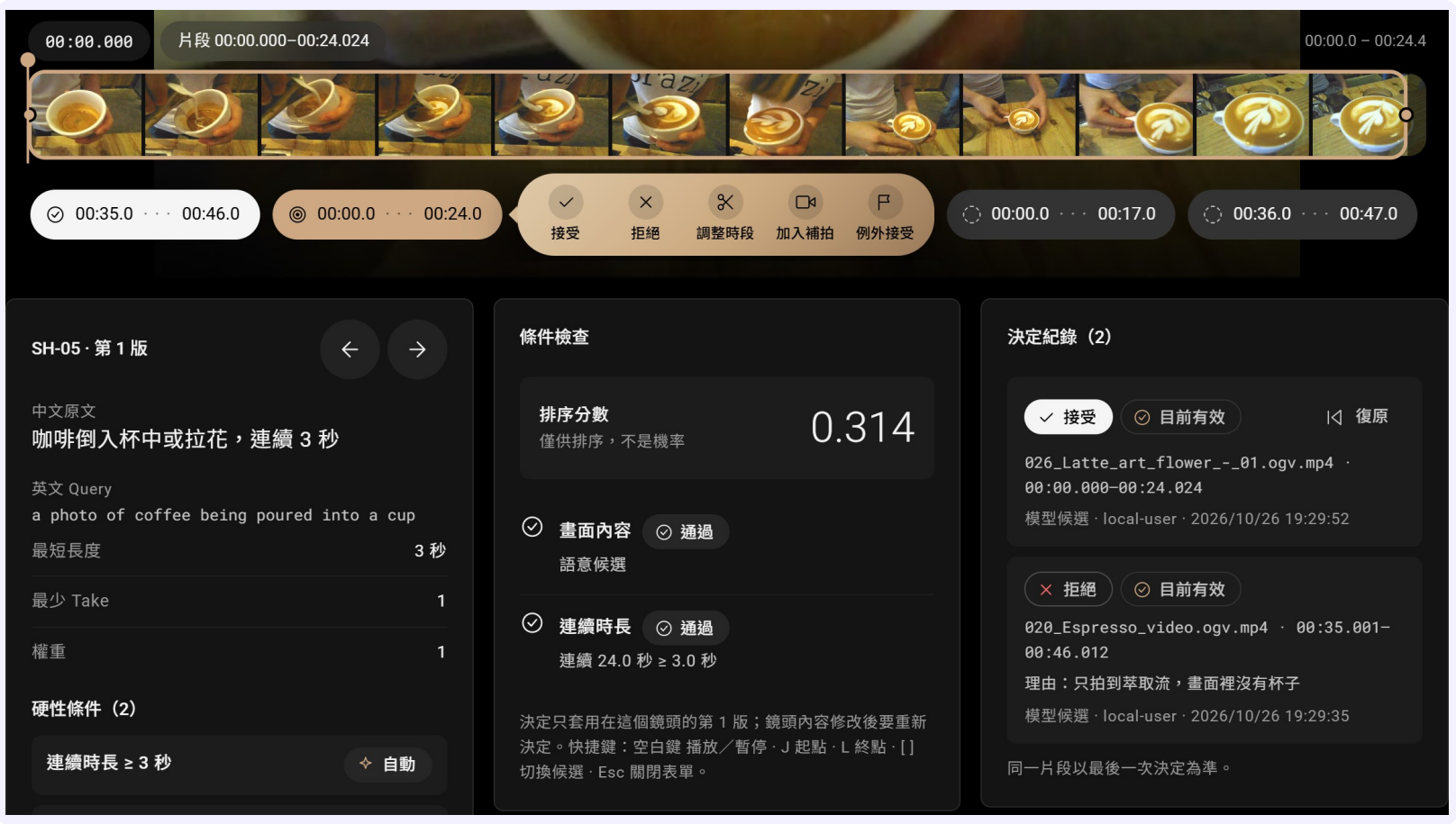
This demo project

- 9 Commons clips, 5.9 min, analysed in 22 s on a desktop GPU
- AI candidate: 10 of 11 shots
- Accepted by a person: 5 of 11 (SH-11 needs a second take)
- Partial: SH-10 (manual brand check)
- Reshoot list: grinder, drinking, coffee bag
- Decisions entered as a scripted demo review, not timed

Live app (Coverage page, cropped). Columns: shot · AI answer · person's decision · reason · clip.

SOLUTION

Every answer plays its clip, and a person decides



SH-05 on this screen

- Shot 「咖啡倒入杯中或拉花，連續 3 秒」 as fixed before any frame was seen (eval/requirements.json, 10/26 18:09)
- Filmstrip and clip ranges on the original timecode
- Ranking score 0.314: orders clips, not a probability
- Length check 24.0 s ≥ 3 s: how long frames stay close to the query
- Log: espresso stream with no cup rejected with a reason; latte-art pour accepted; each can be undone

Live app (clip view, cropped). Original files are never touched.

SOLUTION

Shots without a usable clip go on the reshoot list



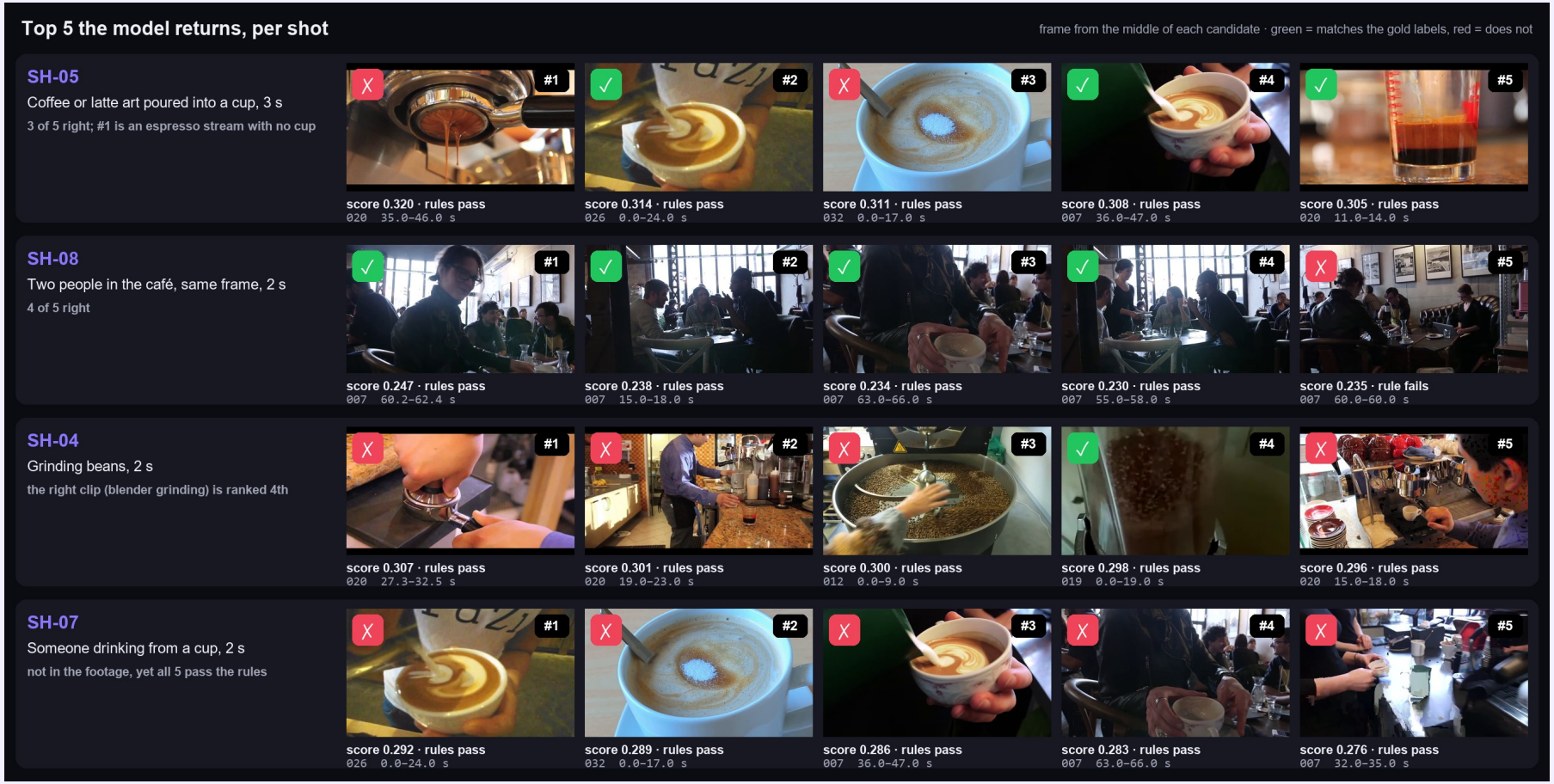
What the crew gets

- One card per shot: why it is missing, the closest clip, minimum length, how it will be checked
- The card shows the AI's rules passed; the person decided the clip shows tamping, not grinding
- Status: to decide → reshoot → shooting → done / not reshooting
- Video demo: the reshot clip was analysed alone in 2.5 s and the grinder shot left the list
- The demo review stopped at the top 3; the 4th candidate was a valid grinding clip (next page)
- Fixed: the card now shows how many candidates a person checked (已檢查 3 ／ 5) and warns when some are unchecked

Live app (reshoot list, SH-04 card). Export: CSV / HTML.

EVIDENCE

What the model returns: top 5 per shot, checked against labels



Read it as

- SH-05: right clips are in the list, but #1 is wrong
- SH-08: same-frame rule works here (4 of 5)
- SH-04: the right clip is 4th; ranking by peak is the fix
- SH-07: nothing to find, yet 5 clips pass, so missing shots go undetected

Demo project (9 clips); candidates and scores from the live app; right / wrong from eval/labels/gold.json (≥ 0.5 s overlap). decks/scenegap/fig_top5.py.

EVIDENCE

How the length rule reads the model's scores



What the curves show

- Street frames score about 0.18; the pour about 0.33, well above τ 0.225
- SceneGap samples 1 frame per second: a 4 s pour gives 4 samples, a lower bound of 3.0 s, so it passes the 3 s rule
- A 2 s pour gives 1.0 s: too short, correctly
- A finished latte scores 0.30 with nothing poured: CLIP sees the result, not the action, so a person checks it
- Stage II: denser sampling near the edges and an action check for pour-type shots

Rule-check cases cut from real Commons footage (eval/rule_check.py); similarity from the same CLIP model at 4 frames/s. decks/scenegap/fig_curves.py.

EVIDENCE

What each rule measures, and what it cannot tell

RULE	WHAT SCENEGAP MEASURES	WHAT IT CANNOT TELL	TODAY
Length	How long frames stay close to the query inside one camera cut (lower and upper bound; re-check every 0.25 s)	Whether the action itself lasts that long; the lower bound runs about 1 s short (4 s pour → 3.0 s)	auto, a person checks
Same frame	Both phrases above the threshold in the same sampled frame	Who is with whom; objects too small for CLIP	auto, a person checks
Takes	Distinct files, by content hash; duplicates of one file are one take	A re-encoded copy of the same take	auto, a person checks
Readable brand	Nothing automatic	Reading text	manual
Not found	Nothing reached the threshold in the analysed files	Proof the shot was never filmed; failed files make it 'can't tell'	auto, scope shown

> **Dense sampling gives more observations, not certainty: a clip that looks like pouring is still checked by a person.**

EVIDENCE

Rule check: 10 of 12 right; both misses are false acceptances

CASE (REAL COMMONS FOOTAGE, CUT TO A KNOWN ANSWER)	SHOULD	SCENEGAP	MEASURED RUN (S)	FILES / TAKES	RIGHT?
SH-05 · continuous 4 s latte-art pour	pass	pass	3.0	1 / 1	yes
SH-05 · 6 s pour from a different shoot	pass	pass	5.0	1 / 1	yes
SH-05 · pour lasts only 2 s (rule: 3 s)	not pass	not pass (duration_short)	1.0	1 / 0	yes
SH-05 · two 2 s pours split by another shot	not pass	not pass (duration_short)	1.0	1 / 0	yes
SH-05 · finished latte, nothing is poured (5 s)	not pass	pass	4.0	1 / 1	NO
SH-08 · two people at a café table, same frame	pass	pass	3.0	1 / 1	yes
SH-08 · two people, but outdoors (not in a café)	not pass	not pass (no_candidate)	—	1 / 0	yes
SH-08 · two people, then a café interior: separate shots	not pass	not pass (no_candidate)	—	1 / 0	yes
SH-11 · one barista clip (rule: 2 takes)	not pass	not pass (takes_short)	4.0	1 / 1	yes
SH-11 · the same clip copied byte for byte	not pass	not pass (takes_short)	4.0	2 / 1	yes
SH-11 · the same clip re-encoded (still one take)	not pass	pass	4.0	2 / 2	NO
SH-11 · two different barista clips	pass	pass	4.0	2 / 2	yes

Open: an action check (a finished latte passes as 'pouring') and re-encoded-take dedup. Such passes stay 'candidate, a person confirms', never done. eval/rule_check.py, frozen threshold.

EVIDENCE

How we tested: fixed protocol, open footage, AI-assisted labels

Protocol and the 11 shots fixed before any label; threshold set on dev only; test scored once. Every file below is in the repo.

Footage

ITEM	VALUE
Source	44 Wikimedia Commons clips, 25.2 min, H.264 ≤ 1280 px
Dev	18 clips, 11.2 min: threshold only
Test	26 clips, 14.0 min: scored once
Distractors	17 of 44 clips contain none of the 11 shots
Licences	CC BY-SA 19, CC BY 14, CC0 8, public domain 3
Split note	one author pair crosses (022 dev, 025 test); without the 2 projects holding 025, SceneGap still catches 0/31

Labels

ITEM	VALUE
Labellers	two independent AI sessions on 1 fps sheets; an AI third pass
Agreement	472/484 clip × shot pairs (97.5%); κ 0.914 over 16,687 frame samples
Third pass	143 disputed frame labels decided with 9 written rules
Consistency pass	78 of 238 agreed positives removed after labelling (65 were SH-11)
Lenient labels	restoring them leaves SceneGap unchanged (7/11, 18/55)
Threshold	τ 0.225: dev F1 0.240, frame precision 14.4%, recall 71.8%

> **Simulated projects reuse the same clips, so they are not independent shoots. Human-checked labels on the original video come next; details in A5–A7.**

EVIDENCE

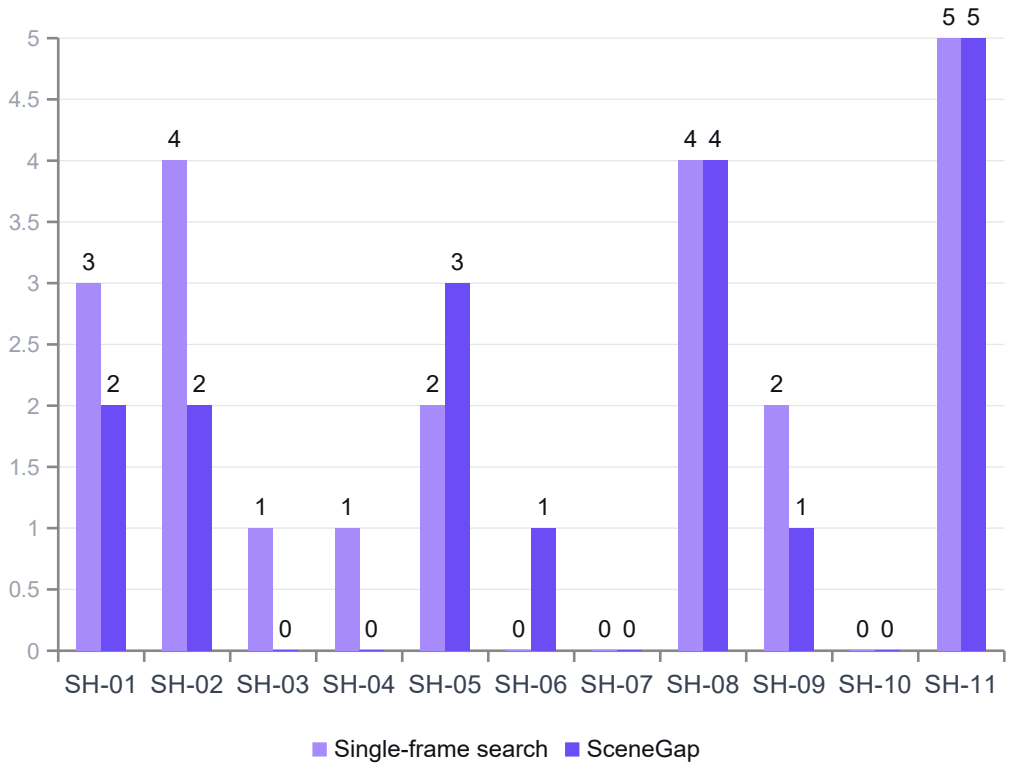
Right clip in the top 5 for 7 of 11 shots; plain frame search found 8

METHOD	RIGHT CLIP IN TOP 5	TOP-5 PICKS RIGHT
Title keywords (whole clips)	45% (5/11, CI 21–72%)	36% (8/22, < 5 returned)
Single-frame search	73% (8/11, CI 43–90%)	40% (22/55)
SceneGap, ranked after rule checks	**64% (7/11, CI 35–85%)**	**33% (18/55)**

Why one hit was lost

- Lost SH-03 and SH-04, gained SH-06.
- Clip 019 is the right answer for both. On its own it passes every rule (5 s and 19 s runs): the rules did not reject it.
- Runs are ranked by average similarity; single-frame search uses the peak frame. Stage II ranks by peak.
- A hit needs 0.5 s overlap, so this layer does not test the length rule (slide 12 does).

Correct clips in the top 5, per shot (of 5)



26 held-out clips, 14.0 min, scored once at τ 0.225; Wilson 95% intervals. SH-03 and SH-07 each rest on one 3 s gold clip. Per shot: A5.

EVIDENCE

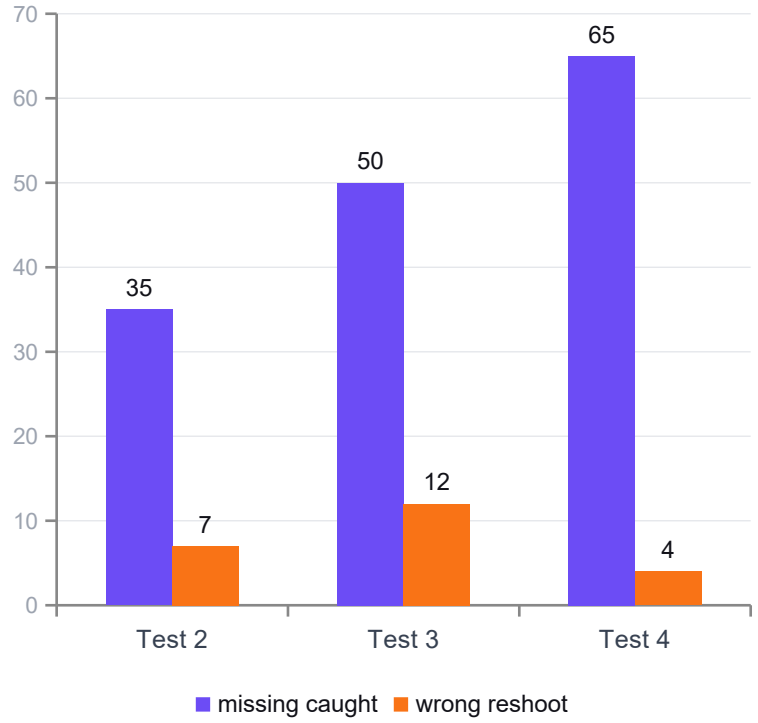
Missing shots: on test 4 the check layer catches 65% at 3.8% wrong reshoots; both targets met

Each test set scored once; targets: caught ≥ 30%, wrong reshoot ≤ 5%

DATA	METHOD	MISSING CAUGHT	WRONG RESHOOT
Test 1 (Oct 26)	system alone	0/38 (0%)	0/50 (0%)
Test 2: 24 new clips	check of the top 3 candidates	32/91 (35%)	3/41 (7.3%)
Test 3: 39 new clips	check of every second (frozen)	76/152 (50%)	8/68 (12%)
Test 3, post hoc	+ 3-frame look (simulated)	76/152 (50%)	2/68 (2.9%)
Test 4: 150 new photo clips	**same check, frozen**	**146/226 (65%)**	**4/104 (3.8%)**
Test 4, preregistered	+ 3-frame look (simulated)	146/226 (65%)	0/104 (0%)

- How: Qwen2-VL-2B answers each shot's check question ('Is ground coffee coming out of a grinder?') for every 1 s frame; 2 confirmed seconds in a row = present.
- Test 3's 8 wrong reshoots were borderline frames, so each alert shows the 3 most likely frames; test 4 then scored that look as a preregistered metric, simulated from the labels (assumes the person recognises the shot; not a user test).
- Test 4 is Commons photos as 4 s clips: fresh images, but no motion or cuts. Upper 95% bound on wrong reshoots: 9.5%.

New footage, % of shots

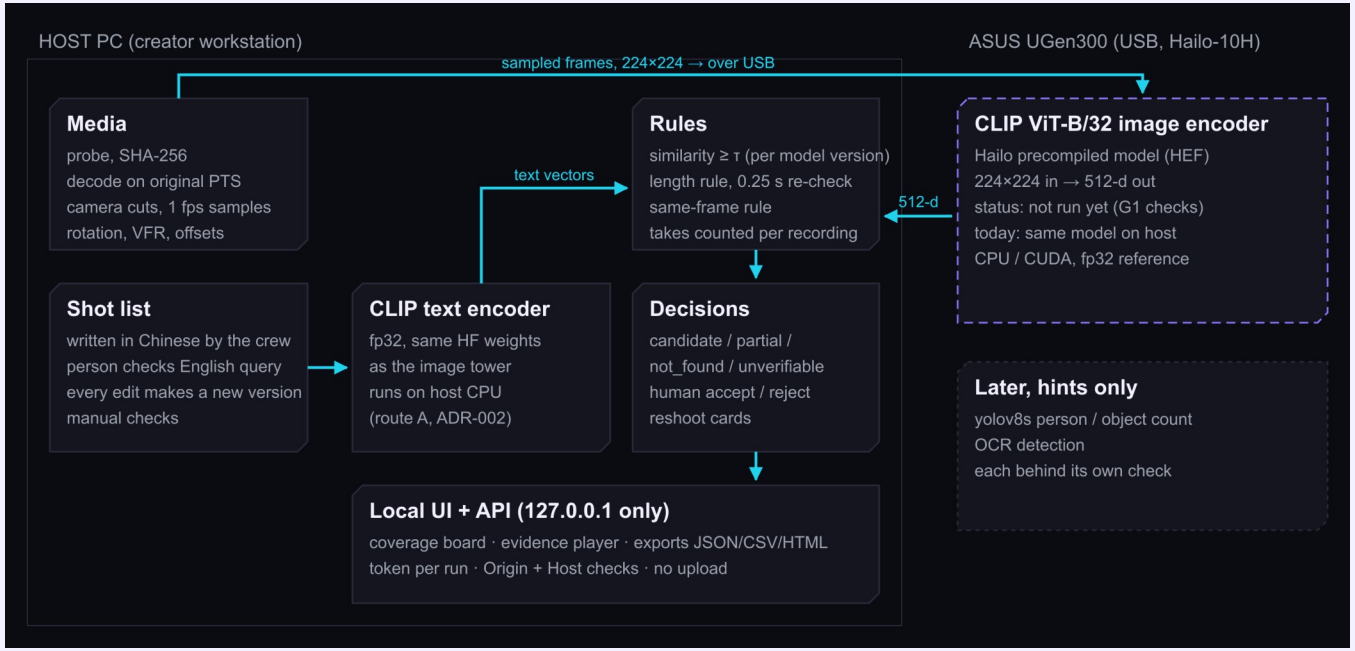


Labels: two AI passes and a blind third (κ 0.93–0.98), not yet checked by people. Next: real café video and a timed user test.

Unit: one shot in one simulated 8-clip project. eval/PROTOCOL.md (Stage 1.5–1.7); eval/verify16.py; eval/verify16_evidence.py; A8.

HARDWARE

Plan: the image half of CLIP on the UGen300; the PC keeps the rest



Second device job: the missing-shot verifier, Qwen2-VL-2B, which Hailo lists for Hailo-10H (slide 15). Frames over USB: about 9 MB per minute (estimate). Both models must pass G1-style checks against the PC first (A1). Demo PC: Windows 11, RTX 4080; Mac: CPU only.

ASUS UGen300

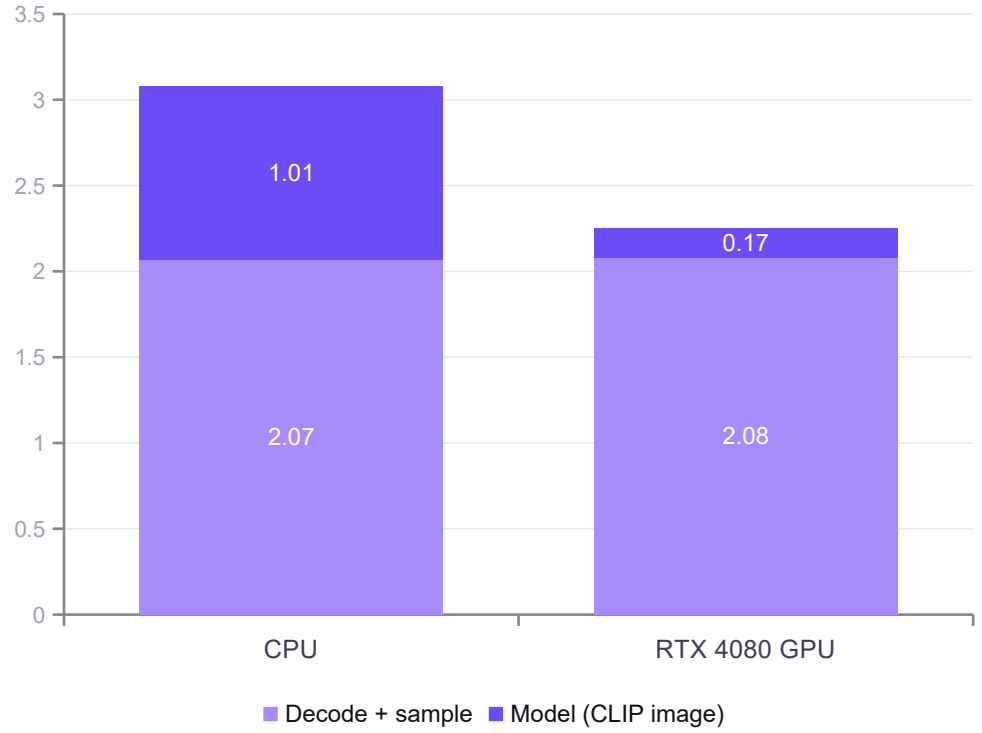
ITEM	VALUE
Chip	Hailo-10H
AI performance	40 TOPS INT4 / 20 TOPS INT8
Memory	8 GB LPDDR4, 4266 MT/s
Link	USB 3.1 Gen2 Type-C, 10 Gbps
Power / size	2.5 W typical; 105 × 50 × 18 mm, 150 g
OS	Windows, Linux, Android (no macOS listed)
Price	$299.99 USB 8 GB (reported)
CLIP ViT-B/32 image encoder	81.5 / 350 FPS at batch 1 / 8 (Hailo zoo, PCIe)

BUSINESS_FACTS.md §2 (ASUS spec; price reported by CNX Software), MODEL_PLAN.md §2, §6; Hailo Model Zoo measures over PCIe, not USB.

HARDWARE

Measured: decoding is 2.1 of the 2.3 s per minute; a UGen300 would free the GPU or a laptop CPU

CLIP index, s per minute of video (check layer not timed)



Model and search

BACKEND	MODEL SPEED	MODEL LOAD	SEARCH P50 / P95 (10,000 SEGMENTS)
CPU	61.5 frames/s	6.51 s	11.77 / 14.29 ms
RTX 4080	365 frames/s	1.88 s	7.61 / 13.39 ms
UGen300 (vendor)	81.5 / 350 FPS (batch 1 / 8, PCIe)	not run	not run

Playing a 1080p30 test pattern while analysing (budget 33.3 ms)

ANALYSIS	LATE FRAMES	FRAME WORK P95	ANALYSES FINISHED
none	0 / 1,800	5.7 ms	0
CPU	0 / 1,800	7.5 ms	22
GPU	0 / 1,800	6.2 ms	52

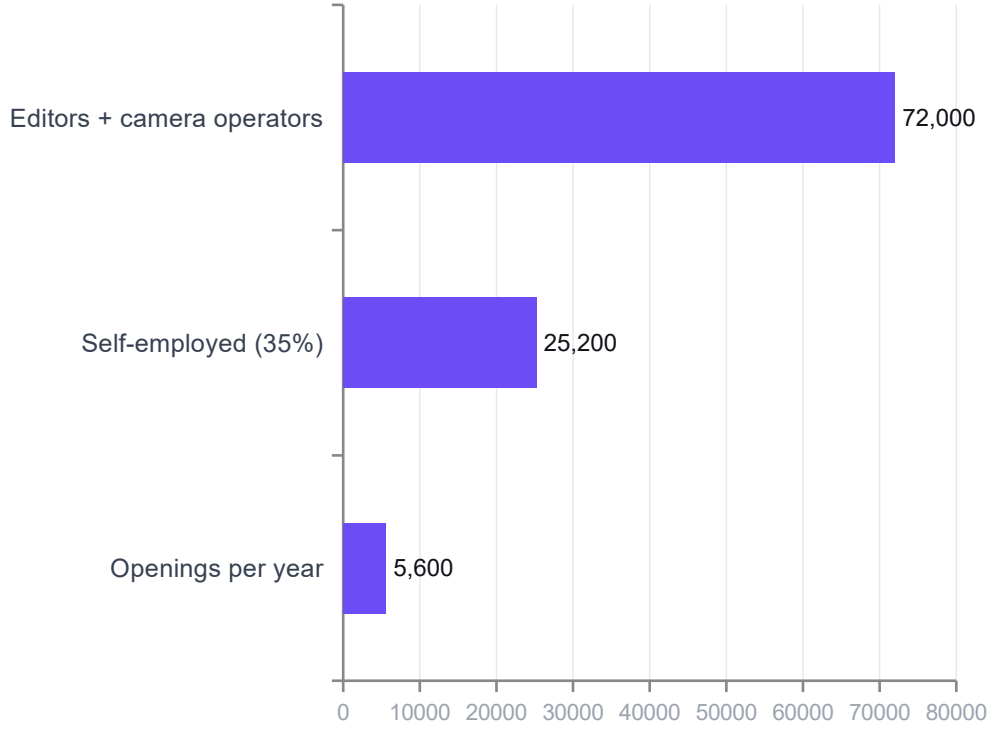
The model is 32% of the CPU run and 7% of the GPU run. Stage II measures an editing app alone vs with CPU, GPU or UGen300 analysis.

eval/results/perf.json (six short clips, 85 s, ≤ 1280 px, warm model; import and hashing not timed), interference.json (synthetic test pattern, not an editing app).

BUSINESS

Sourced: who pays first is 1–3 person crews and freelance editors

US floor (BLS OOH, 2025)



Segments (PRD 4.1)

SEGMENT	SIZE SIGNAL	MUST HOLD TRUE
1–3 person crews (first)	≈ 11,400 self-employed US camera operators; TW crew-day NT$12,000 –28,000	they write a shot list and carry a computer on site
Freelance editors (first)	≈ 13,800 self-employed US editors; median $75,420/yr	client briefs are specific; a pickup can still be arranged
Student teams	no data	will check a little by hand
Brand content producers	91% of businesses use video (Wyzowl 2026, partial)	templates stay stable

BLS is a floor: it leaves out YouTubers and in-house marketing staff. Worldwide, ~50M creators and ~4% professional (Goldman Sachs, partly verified).

US BLS OOH accessed 2026-10-26 (39,400 editors + 32,600 camera operators; 35% self-employed; ~5,600 openings a year); self-employed split is our arithmetic. Day rates per videographer.

BUSINESS

Hypothesis: one prevented reshoot could offset the device cost

Value = hours saved × hourly cost + reshoot avoided − tool cost

TERM	VALUE USED	STATUS
Hourly cost, editor	$36.26/h = $75,420 ÷ 2,080 h	sourced; 2,080 h assumed
Reshoot day, US, 2 people	$1,600–3,000 = 2 × $800–1,500	sourced rates; crew of 2 assumed
Reshoot day, Taiwan	NT$12,000–28,000 per crew-day	quote statistics
Tool cost	$299.99 UGen300; core software free	price reported
Hours saved	unmeasured (proxy: 38% of footage to play)	user study, Stage II
Reshoots avoided per project	unmeasured	interviews, Stage II

8.3 h
of editor time pays for the device

0.10–0.19
of one US two-person reshoot day

Price per seat, USD per year or one-time



Open core: the Apache-2.0 checker is free. Paid: a tested UGen300 bundle (installer, driver versions, device-checked model, support) and shot-list templates. Axle FOCUS: $2,995+/yr.

Device price only: the on-site computer, setup, upkeep and the person's time are extra. Test it: two groups check one card, by hand vs with SceneGap.

BUSINESS_FACTS.md §1c, §2, §4; * vendor page not readable at access. Yearly = 12 × monthly list price.

PLAN

Shown today, proven next

SceneGap Edge today: shot list → candidate clips with evidence → a person's decision → reshoot list → re-check, on real footage.

PROOF	TODAY	STAGE II TARGET	SOURCE
1 · Rules	10 of 12 controlled cases; top 5 right for 7/11; missing shots: test 4 caught 65% at 3.8% wrong reshoots (0% with a simulated 3-frame look; photo clips)	Recall@5 ≥ 90% and wrong reshoot ≤ 5% in the supported scope; ranking by peak; per-shot thresholds	PRD 21.5
2 · Labels	AI-assisted, κ 0.91	a person labels sampled pairs on the original video	Protocol
3 · People	no study; proxy 38% of footage to play	3–5 interviews, then 8–12 crews; median task time −30%	PRD 22.1
4 · UGen300	not run; PC harness cos 1.0000, ρ 0.99999	G1a cos ≥ 0.98, G1b ρ ≥ 0.95, G1e status ≥ 90%; USB speed, power, editing-app load	MODEL_PLAN §4
5 · Speed	search p95 13–14 ms; CLIP index 2.3 s per minute (GPU); check layer not timed	first result for a 60 s clip ≤ 45 s, timed end to end	PRD NFR-05
6 · Software	52 automated tests; 30 of 36 acceptance cases pass; clean rebuild passes	the device and user cases (TC-017, 018, 033, 034)	A9, A12

> **The crew leaves the location with a reshoot list and the evidence a person checked.**

Code, data and results: github.com/yu9bryan/scenegap-edge (Apache-2.0). References: A13.

APPENDIX

A1 · G1: six checks before the UGen300 replaces the PC model

CHECK	MEASURE (DEV FRAMES, SAME PREPROCESSING)	PASS BAR	STATUS
G1a	Cosine, HEF vs fp32 embedding, per frame	median ≥ 0.98, p5 ≥ 0.95	not run yet
G1b	Spearman ρ of similarity per query	median ≥ 0.95, worst ≥ 0.85	not run yet
G1c	Top-5 / top-10 shot overlap	≥ 0.80 / ≥ 0.85	not run yet
G1d	Dev Hit@5, each backend with its own τ	no dev shot lost vs fp32	not run yet
G1e	Status agreement on the dev project simulation	≥ 90% same status; no new 'candidate' on a true gap	not run yet
G1f	Frames/s over USB, 60 s clip end to end	reported, no bar	not run yet
Harness	Same script, CPU vs CUDA fp32 (662 frames, 13 queries; plan ≥ 2,000 / ≥ 40)	≈ 1.0 expected	cos 1.0000, ρ 0.99999

Vendor: Hailo-10H ViT-B/32 image encoder, CIFAR-100 zero-shot 65.3 float → 64.5 on chip (PCIe, not USB).

docs/MODEL_PLAN.md §2, §4; eval/results/g1_selfcheck.json; Hailo Model Zoo (vendor figures).

APPENDIX

A1b · After G1: product-level checks on the device

CHECK	WHAT IT PROTECTS AGAINST	STATUS
Status agreement, UGen300 vs PC, on dev projects	quantisation causing more false acceptances	not run yet
USB end-to-end analysis time for a 60 s clip	a device that is fast in the zoo but slow over USB	not run yet
Editing-app playback and scrubbing with analysis on the device	claiming host relief without measuring it	not run yet
Whole-system extra power (meter at the wall)	quoting 2.5 W as the system's cost	not run yet
Missing shots caught and wrong reshoots on new footage, verifier on the device	judging the device by matching vectors alone	not run yet

> **Model agreement (G1) is necessary, not sufficient: the product effect is measured on new footage.**

APPENDIX

A2 · What could break, and what we do then

RISK	FALLBACK
HEF fails G1 (quantisation hurts ranking)	Recalibrate with dev frames; TinyCLIP; else ship the CPU/GPU version and say so
A borderline camera cut splits a long take	The length check says 'can't tell', not 'too short'; the cut is shown
One global threshold fits some shots badly	One threshold per model version now; per type of shot next, set on dev
English phrase drifts from the Chinese shot	A person checks the phrase; a multilingual model only after its own check
AI-assisted labels, Commons footage, small test	Intervals shown; Stage II adds human labels and self-shot projects
Crews will not write shot lists	Templates; the interviews check this first

APPENDIX

A3 · Stage II order: rules and people first, then the device

Wk 1–2	Wk 2–3	Wk 3–5	Wk 4–6	Wk 6+
Rules and labels	**Missing shots**	**User study**	**UGen300**	**Chinese queries**
More controlled cases; a person labels sampled pairs on the original video; rank runs by peak.	Per-shot thresholds with a project-level target; preregister, score once on new footage.	3–5 interviews, then 8–12 crews: by hand vs SceneGap, counterbalanced; time, misses, wrong reshoots.	G1 on the device; editing-app load with CPU, GPU or UGen300 analysis.	A multilingual model (SigLIP 2), adopted only if it finds as many dev shots.

APPENDIX

A4 · Footage shown in this deck

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
001	Vichy - Grand Café, terrasse sous la grêle.ogv	Romainbehar	CC0
007	Café Lomi, Paris - Vimeo.webm	ideafarm films	CC BY 3.0
012	Kaffe - Ystad-2024.ogg	Jonn Leffmann	CC BY 3.0
016	Lions Almelo zamelt koffie in voor Voedselbank.webm	AAVISIE	CC BY 3.0
019	MoliendoCafeEnLicuadora ohs02.webm	PetrohsW	CC BY-SA 4.0
020	Espresso video.ogv	Rosatrieu	CC BY-SA 3.0
026	Latte art flower - 01.ogv	Anna Frodesiak	CC0
032	Sugar falling into coffee.webm	Anselm Schüler	CC BY-SA 4.0
041	Empty Streets of Paris COVID19.webm	CreatorsCloud	CC BY 3.0

Screenshots and cover frames come from these clips. All 44 clips used in the evaluation, with links: docs/CREDITS.md.

APPENDIX

A5 · The 11 shots and per-shot results (correct / returned, top 5)

SHOT	IN ENGLISH	RULE	TEST CLIPS WITH GOLD	SCENEGAP	SINGLE-FRAME	KEYWORDS	MISSING IN PROJECTS
SH-01	Storefront from the street	≥ 3 s	3 (36 s)	2/5 ✓	3/5 ✓	1/2 ✓	3 of 8
SH-02	Café interior, tables	≥ 3 s	2 (39 s)	2/5 ✓	4/5 ✓	1/4 ✓	1 of 8
SH-03	Coffee-bean close-up	≥ 2 s	1 (3 s)	0/5	1/5 ✓	0/3	6 of 8
SH-04	Grinding beans	≥ 2 s	2 (18 s)	0/5	1/5 ✓	1/1 ✓	6 of 8
SH-05	Coffee or latte art poured	≥ 3 s	6 (48 s)	3/5 ✓	2/5 ✓	3/4 ✓	1 of 8
SH-06	Cup of coffee on a table	≥ 2 s	8 (46 s)	1/5 ✓	0/5	0/5	0 of 8
SH-07	Someone drinking	≥ 2 s	1 (3 s)	0/5	0/5	0/0	8 of 8
SH-08	Two people, same frame	≥ 2 s, same frame	3 (56 s)	4/5 ✓	4/5 ✓	0/0	1 of 8
SH-09	Espresso extraction	≥ 2 s	2 (30 s)	1/5 ✓	2/5 ✓	2/3 ✓	4 of 8
SH-10	Bean bag, brand readable	≥ 1 s, brand (manual)	2 (4 s)	0/5	0/5	0/0	6 of 8
SH-11	Barista, 2 takes	≥ 3 s, 2 takes	5 (74 s)	5/5 ✓	5/5 ✓	0/0	2 of 8

eval/requirements.json, eval/labels/gold.json, eval/results/test_metrics.json. ✓ = a correct clip in the top 5.

APPENDIX

A6 · Labels: agreement, third pass and the consistency pass

Agreement between the two AI annotators

MEASURE	VALUE
Clip × shot pairs agreeing (IoU ≥ 0.5 or both empty)	472 / 484
Frame samples (1 fps × 11 shots)	16,687
Both positive / A only / B only	808 / 46 / 97
Observed agreement / Cohen's κ	99.1% / 0.914
Disputed labels to third pass	143

Consistency pass (added after labelling, before calibration)

SHOT	AGREED POSITIVES CHECKED	REMOVED	RULE
SH-03	17	4	R6
SH-04	28	9	R8
SH-09	41	0	R2
SH-11	152	65	R3

Rules the third pass wrote (examples)

- R3 barista: commercial setting, actively preparing coffee
- R5 anything poured into the cup → SH-05, not SH-06
- R8 grinding: grounds visibly come out of the spout

> **Only removals were possible; restoring all 78 leaves SceneGap's numbers unchanged (lenient labels, A7).**

eval/labels/agreement.json, consistency.json, adjudication_rules.json.

APPENDIX

A7 · Threshold on dev; lenient labels on test

Dev sweep (7,282 frame × shot pairs, 432 positive)

τ	TP	FP	FN	PRECISION	RECALL	F1
0.205	356	2,627	76	11.9%	82.4%	0.2085
0.210	351	2,374	81	12.9%	81.2%	0.2224
0.215	342	2,160	90	13.7%	79.2%	0.2331
0.220	331	1,992	101	14.2%	76.6%	0.2403
0.225	**310**	**1,837**	**122**	**14.4%**	**71.8%**	**0.2404**
0.230	280	1,671	152	14.4%	64.8%	0.2350
0.235	262	1,511	170	14.8%	60.6%	0.2376
0.240	239	1,370	193	14.9%	55.3%	0.2342
0.245	218	1,247	214	14.9%	50.5%	0.2298

Test, with the 78 removed labels restored

METHOD	HIT@5	PRECISION@5	MISSING CAUGHT
Title keywords	6/11 (55%)	9/22 (41%)	33/37 (89%)
Single-frame	8/11 (73%)	22/55 (40%)	0/37 (0%)
SceneGap	7/11 (64%)	18/55 (33%)	0/37 (0%)

- F1 is flat between 0.220 and 0.225; τ 0.225 chosen by the rule written before labels (ties go higher).
- Precision stays at 12–15% across the range: no single threshold separates shots well frame by frame.
- Centre crop beat padding by 0.0035 F1 (center-crop 0.2404, pad-square 0.2369).

eval/results/calibration.json, test_metrics.json (sensitivity_lenient_gold).

APPENDIX

A8 · Missing-shot trade-off on dev (not used to rescore)

SCORE	SETTING	MISSING CAUGHT	WRONG RESHOOT	'FOUND' WHEN ABSENT
raw cosine	τ 0.225	5/89 (6%)	0/131 (0%)	80/89 (90%)
raw cosine	τ 0.24	13/89 (15%)	7/131 (5%)	71/89 (80%)
raw cosine	τ 0.27	24/89 (27%)	14/131 (11%)	65/89 (73%)
raw cosine	τ 0.285	32/89 (36%)	22/131 (17%)	49/89 (55%)
raw cosine	τ 0.29	41/89 (46%)	40/131 (31%)	37/89 (42%)
raw cosine	τ 0.3	46/89 (52%)	62/131 (47%)	37/89 (42%)
margin over generic prompts	0.0	6/89 (7%)	0/131 (0%)	76/89 (85%)
margin over generic prompts	0.0075	11/89 (12%)	0/131 (0%)	70/89 (79%)
margin over generic prompts	0.02	16/89 (18%)	7/131 (5%)	66/89 (74%)
margin over generic prompts	0.0475	29/89 (33%)	14/131 (11%)	51/89 (57%)
margin over generic prompts	0.05	30/89 (34%)	14/131 (11%)	51/89 (57%)

eval/results/explore_dev_gap_tradeoff.json: 20 dev projects × 8 clips. Margin = similarity minus the best of 6 generic prompts.

APPENDIX

A9 · Acceptance cases: 30 of 36 pass by automated check

STATUS	COUNT	CASES (TC-)
pass	30	001, 002, 003, 004, 005, 006, 007, 008, 009, 010, 011, 012, 013, 014, 016, 019, 020, 021, 022, 023, 024, 025, 026, 027, 028, 029, 030, 031, 035, 036
not applicable	1	015
not run	1	017
partial	1	018
measured	1	032
proxy	1	033
manual check	1	034

52 automated tests; TC-035 is the clean rebuild from the public package index.

eval/results/ACCEPTANCE.md (each case with the test that covers it), clean_env.json.

APPENDIX

A10 · Hailo-10H model zoo: options for the device

MODEL (HEF)	ACCURACY, FLOAT → ON CHIP	FPS BATCH 1 / 8	ROLE
clip_vit_b_32_image_encoder	CIFAR-100 zero-shot 65.3 → 64.5	81.5 / 350	planned: same OpenAI weights as the PC model
clip_vit_b_32_text_encoder	Retrieval@10 90.6 → 89.3	48.9 / 98.8	not planned: text stays on the PC
tinyclip_vit_61m_32 image	72.8 → 69.8	101 / 388	fallback if G1 fails (new threshold)
siglip2_b_32_256 image	74.7 → 70.4	64.7 / 221	multilingual option for Chinese shot lists
clip_vit_b_16 image	68.6 → 64.6	51.5 / 158	finer detail, slower
Qwen2-VL-2B-Instruct (VLM)	listed for Hailo-10H (GenAI zoo)	see vendor page	planned: the missing-shot verifier

> **At batch 8 the vendor figure (350 FPS) is close to the RTX 4080 we measured (365 frames/s); the device's case is freeing the host, not raw speed.**

Hailo Model Zoo, Hailo-10H, PCIe Gen3 ×4 host (not USB); docs/MODEL_PLAN.md §2.

APPENDIX

A11 · Market and price sources

FACT	VALUE	SOURCE	VERIFIED
US film and video editors / camera operators	39,400 / 32,600 (2025); 35% self-employed	US BLS OOH, accessed 2026-10-26	yes
Median pay, editors / camera operators	$75,420 / $74,990 (May 2025)	US BLS OOH	yes
Day rate, mid-level videographer (US)	$800–1,500	Film Lifestyle rate guide	yes (estimate)
Event video, full day (Taiwan)	NT$12,000–28,000	PRO360 quote statistics	yes (estimate)
UGen300 USB 8 GB	$299.99 (M.2 $259.99)	CNX Software; ASUS spec page	partial
Premiere Pro single app / Jumper	$22.99/mo / from $49/mo	Adobe checkout excerpt / Jumper site	no / partial
Resolve Studio / Final Cut Pro	$295 / $299.99 one-time	Blackmagic / Apple	yes
Frame.io Pro / Team; Creator Studio	$15 / $25 per member per month; $129/yr	Frame.io / Apple	yes
Axle AI FOCUS	$2,995–28,995 per year	trade press (IBC 2026)	yes
Creators worldwide	~50M, ~4% professional	Goldman Sachs	partial

docs/research/BUSINESS_FACTS.md (URLs, access dates and verified flags for every row).

APPENDIX

A12 · Engineering: 52 tests; 30 of 36 acceptance cases pass

These show the software is reliable, not that the AI is accurate. Other 6: 1 not applicable, 1 not run, 1 partial, 1 measured, 1 proxy, 1 manual check (needs a UGen300, a detector, or users).

Media edge cases

Rotation, 29.97 fps, VFR, start offsets, corrupt and unsupported codecs, files still copying, moved files found again by hash.

Logic

'Can't tell' stays 'can't tell'; two short clips never count as one long one; duplicates are not two takes; editing a shot voids old decisions.

Security

127.0.0.1 only, per-run token, Origin + Host checks, authorised folders, CSV formula guard, escaped HTML, errors never expose internals.

Offline

Core flow tested with outbound sockets blocked. Footage never leaves the computer.

Failure

Disk full stops safely and resumes; interrupted jobs resume; originals' hashes unchanged.

Rebuild

A fresh venv from the public index passes the test suite; Dockerfile and CI included; Apache-2.0.

eval/results/ACCEPTANCE.md lists all 36 cases with the test that covers each (A9).

APPENDIX

A13 · References and source code

ITEM	REFERENCE	LINK
Source code (Apache-2.0)	SceneGap Edge repository; release v0.1.0 has this deck and the video	github.com/yu9bryan/scenegap-edge
Project page	Summary, results and footage credits	yu9bryan.github.io/scenegap-edge
CLIP ViT-B/32	Radford et al., Learning Transferable Visual Models From Natural Language Supervision, ICML 2021	huggingface.co/openai/clip-vit-base-patch32
Qwen2-VL-2B-Instruct	Wang et al., Qwen2-VL, arXiv:2409.12191, 2024	huggingface.co/Qwen/Qwen2-VL-2B-Instruct
ASUS UGen300	ASUS UGen300 USB 8 GB tech spec	asus.com (UGen300 USB 8G tech spec)
Hailo-10H models	Hailo Model Zoo, HAILO10H zero-shot classification; GenAI model list	github.com/hailo-ai/hailo_model_zoo; hailo_model_zoo_genai
Footage	Wikimedia Commons clips and photos, CC0 / PD / CC BY / CC BY-SA	datasets/manifests/*.json in the repo
Statistics	Wilson score interval (Wilson 1927, JASA); Cohen's kappa (Cohen 1960)	eval/PROTOCOL.md
Market and competitors	US BLS OOH; vendor documentation and price pages (A11)	docs/research/BUSINESS_FACTS.md
Device plan sources	[S1]–[S20]: Hailo zoo configs, hailo-apps, community threads	docs/MODEL_PLAN.md, Sources

Full URLs and access dates are in the files named in the last column.